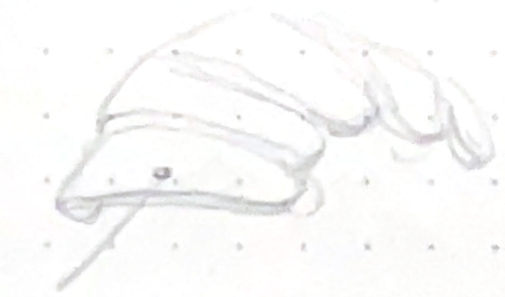
plushie + ball



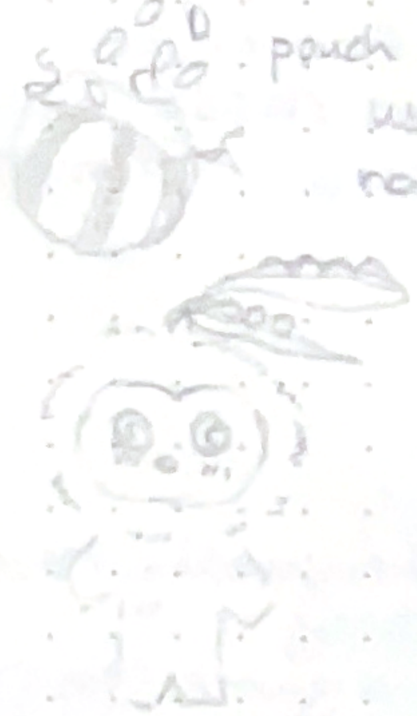
change color

pull to roll

straps creatable
connects to vegetable plush

wrapped in cloth

interactive narrative



pouch from a plushie that gives
user pouch seed based on a
narrative in a book/video

don chorus

2

- hard + soft
- plushie + ball

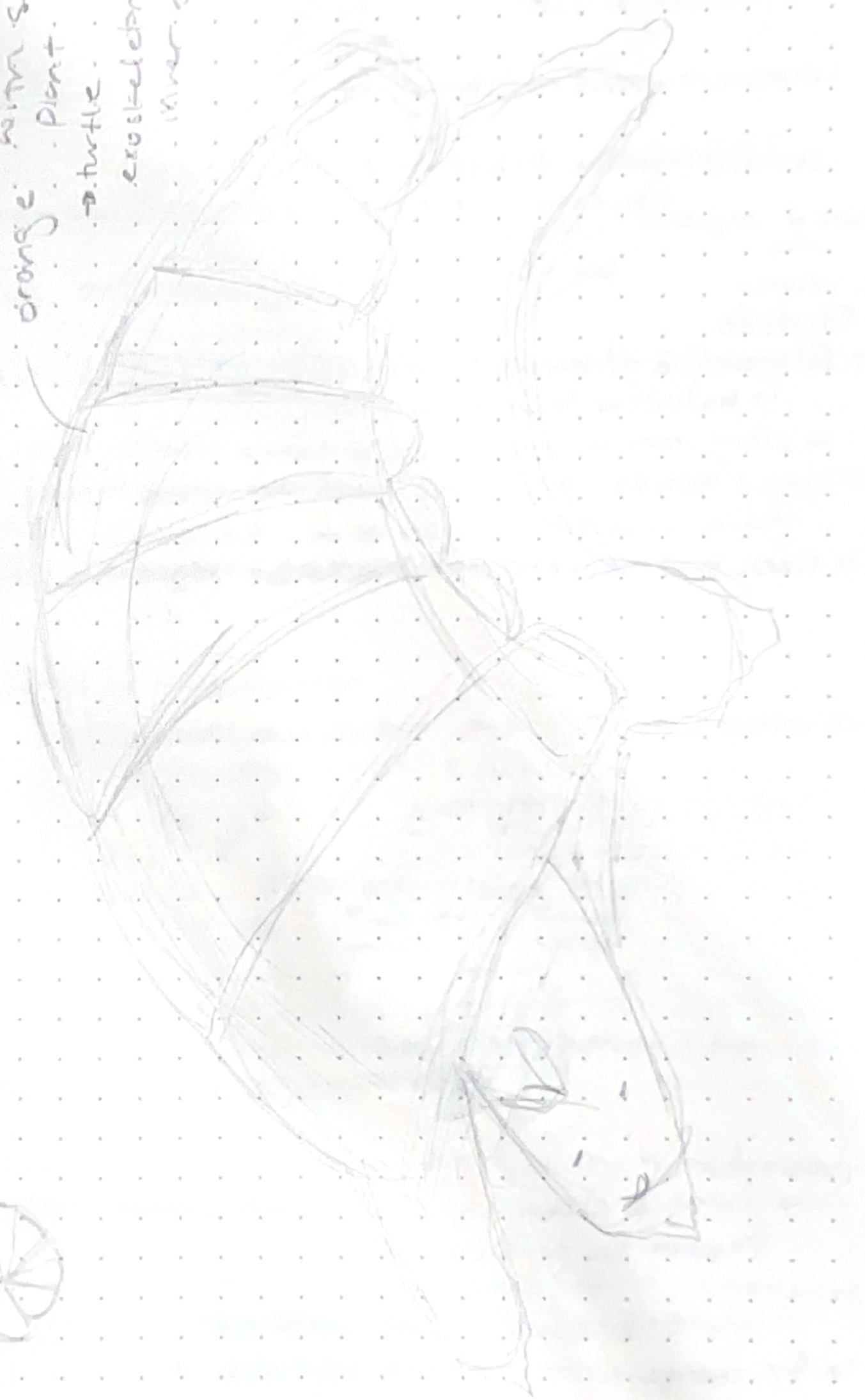
tiny animal with
more direct relationship

orange: with seed or
Plant
→ turtle

exoskeleton-plant

inner core

2 hours



3

Seed companies: baker creek seeds

Dorelle

- color changing = when the plant
- storytelling + USB
- music → wind rattling → moisture sensor of when to water plant (plays music)

seeds + native (sweet + something to eat)
children - less patience 25.-30 days goal!

Michelle

① stacking of birds + bird seeds → lego inspiration puzzle
↳ balance learning (kindergarden)
↳ stem learning of birds?
↳ seeds pollination, seed are transferred - narrative from website
↳ check out variety of colors for vegetable + fruit

Brightly

V1
- robotic toy + potty tray

Kid → robot → plant (8-12 yr old)
water, water — seed: flower

V2
- mix seed + water + fertilizer
↳ scatter in garden (outside)

problems
↳ furry → cleaning? → pet = attachement
↳ roam in garden → dirty

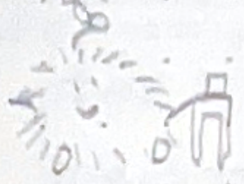
solution
↳ robotics need to be waterproof
↳ fur change color: fabric color change to neut

①

holds chalk
or

light source with circular cutton

- seat + UI: wood
sit on chair = light up
change color
light
chalk + light mobility
the color changed based on light/chalk
chalk/light
image only appears on web with different color (green chalk = green color on web)

② robot pet that gardens

- robot pet that finds perfect soil conditions to plant mixture
mixture = seed + water + fertilizer

③ hostile design + rainwater
- collect rainwater and when someone sits on it, it sprays water

can't see ur

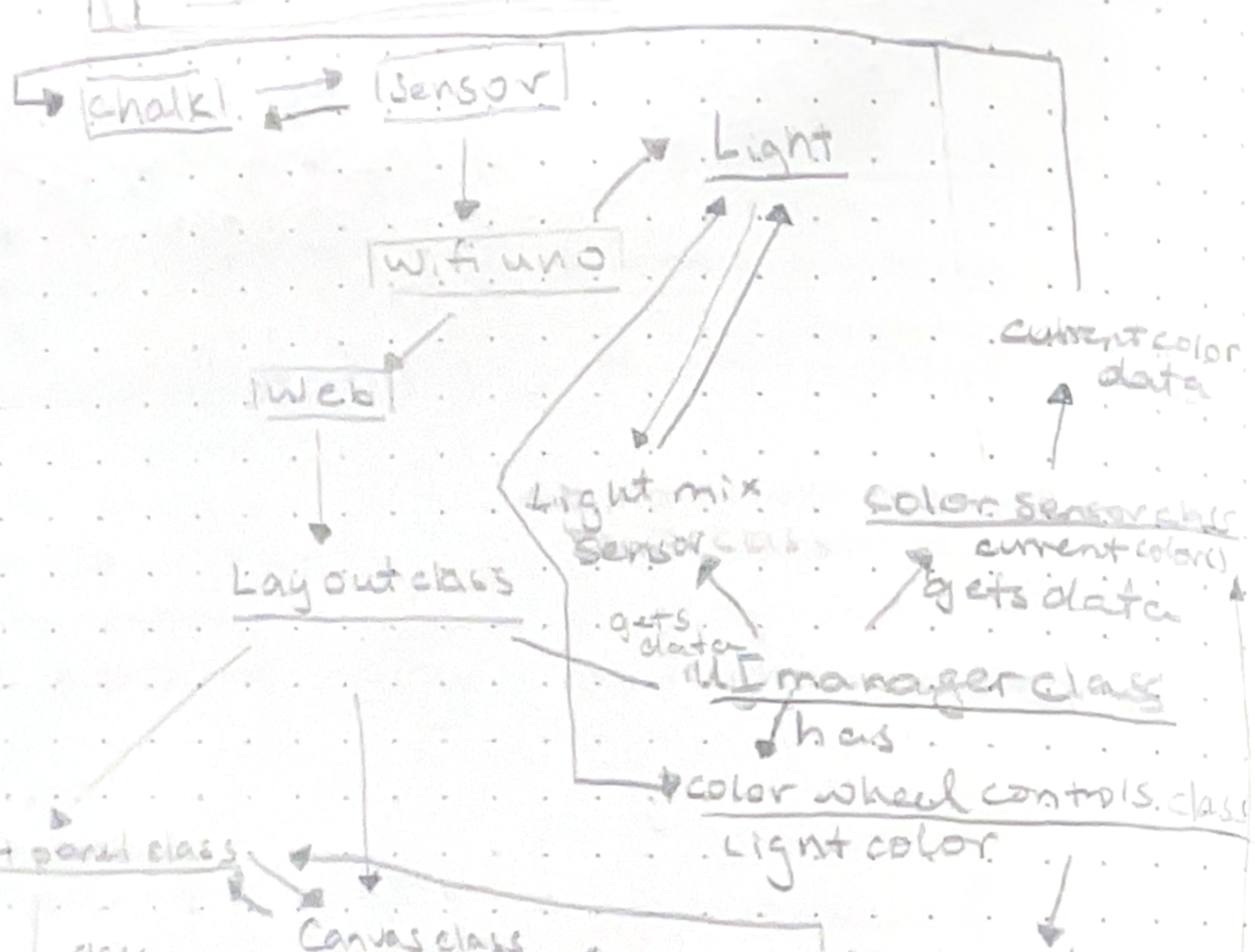
the holes are very visible = warning

④ humidifier that only humidify with hand underneath

⑤
- alarms if a person walks

① assuming sensor detect chalk color, light option = color current color mixing

color wheel



- raspberry Pi camera

③ Laminar flow, touch designer

HW1 - RREF, No solution, gaussian elimination

coefficient matrix = A

e.g $\begin{cases} 2x_1 + 3x_2 = 4 \\ 4x_1 + x_2 = 3 \\ 5x_1 + 6x_2 = 1 \end{cases}$

coefficient matrix

$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \\ 5 & 6 \end{bmatrix}$

augmented matrix $[A|b]$

$A = \left[\begin{array}{cc|c} 2 & 3 & 4 \\ 4 & 1 & 3 \\ 5 & 6 & 1 \end{array}\right]$

vector equation

$\begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} x_1 + \begin{bmatrix} 3 \\ 1 \\ 6 \end{bmatrix} x_2 = \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$

HW2

The equation $A\vec{x} = \vec{b}$ has a solution if $\vec{b}$ is a linear combination of columns of A.

→ see Nfan Lecture 5. pg 56

$\{ \vec{b} : \vec{b}_1 - \frac{1}{2}\vec{b}_2 + \vec{b}_3 = 0 \}$

Reduced Row echelon form

① a pivot in each row.
② if a free variable the steps that column and continue the pivot → normal
③ if matrix does not consist of all 0 then first nonzero # of row = 1.
④ final row should be all 0 or ... cannot be reduced further.

pivot position = the diagonal 1.
free variable = not part of diagonal 1
basic var = # under pivot is.

no solution

if there is row of 0 in last row and b = nonzero then system is inconsistent and 0 = nonzero is impossible = no solution

b = 1 or b = 100 ... impossible
0 = 1 0 = 100

singular value decomposition

$$A = U \Sigma V^T$$

① Compute A^TA
② orthogonally diagonalize A^TA
- eigenvalue + eigenvectors

if not orthogonal use gram-schmitt
- orthonormal + orthogonal

③ find rank ← # of non zero eigenvalues of A^TA

$V = [v_1\ v_2\ v_3]$ → 3×3 matrix

$\Sigma = [2 \times 3 \text{ matrix}]$

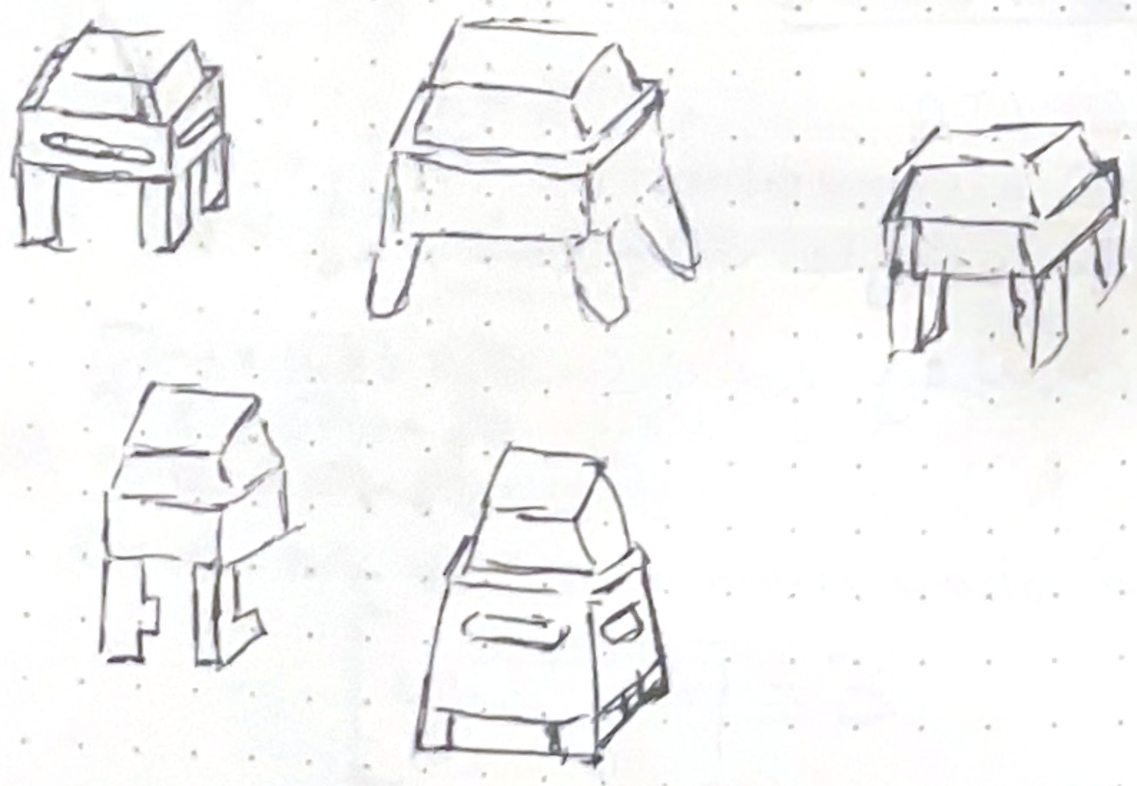
④ normalize $\{Av_1 .. Av_n\}$ to get an orthonormal basis of $\text{Col}\,A$

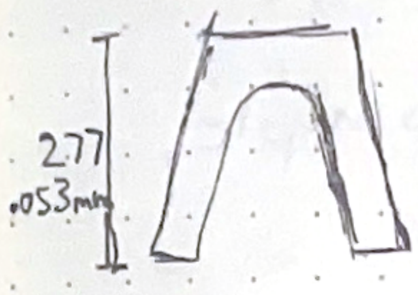
$$u_1 = \frac{Av_1}{\sigma_1} \rightarrow u_1 = \frac{A^TAv_1}{\sigma_1}$$

$\{v_1 .. v_m\}$

⑤ extend to a basis of R^m

⑥ $A = U \Sigma V^T$





- Startup idea : person

Nicholas

questions to ask tees

- if you are printing multiple on 3D print from Rhino, export

me turn →
- make canvas bigger, have a save button
- UI better
- children's small seam
- children rough, rubber
- Daniel suggest screen on side of chair instead of light which will create the pattern

questions

marked

ects

harry
- sense of mystery and reaction
- feels alive

Scrapy
- sound piece with visual representation of birds

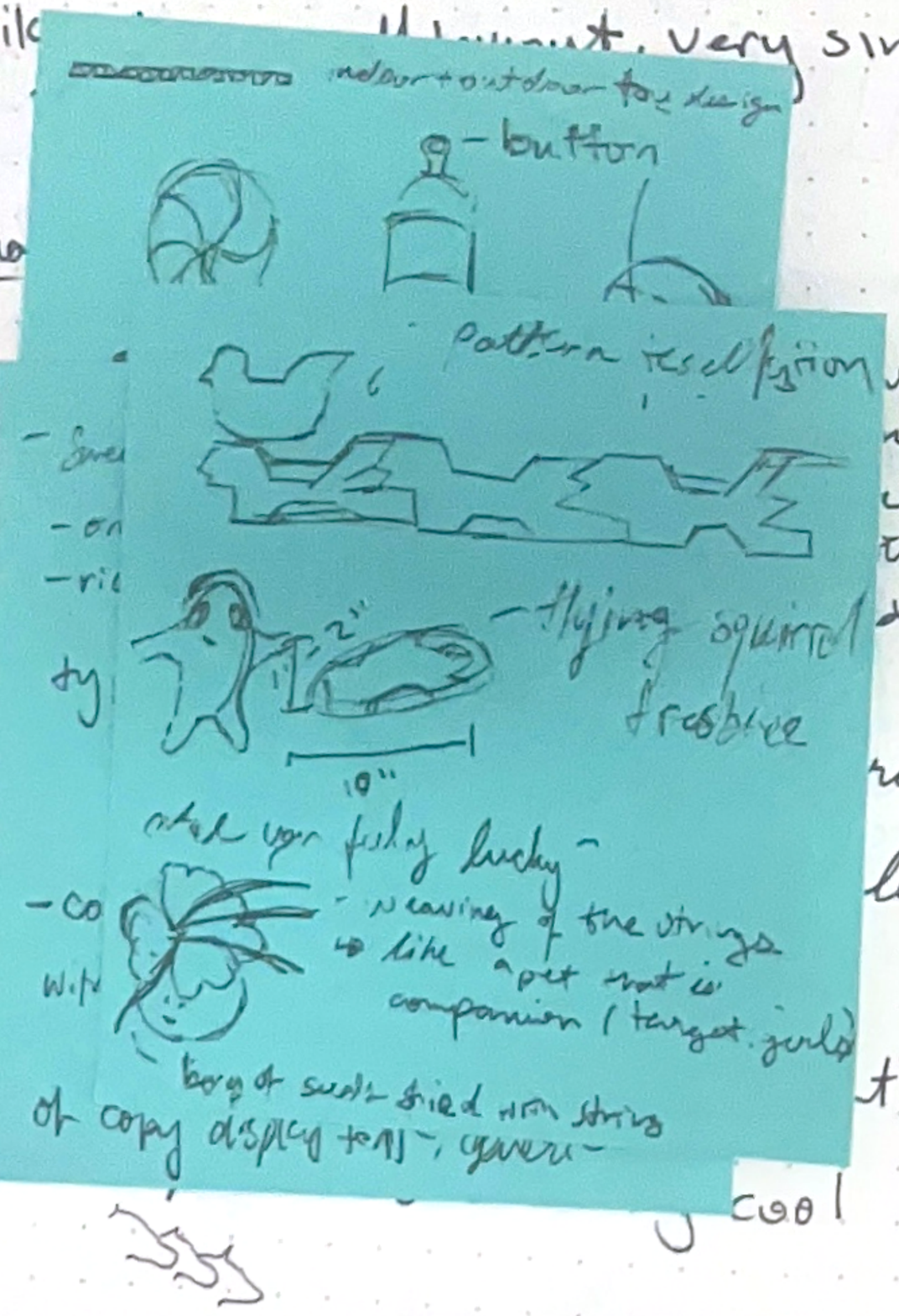
kylar - breathing

Core 2 Crit drawn May 21 2026

- chat with a bot & describe your sound, then based on what you describe it will play sound. - download file and on the side you the sound mode will appear
- also library of different sounds (detailed is copy and paste

next step: if you can directly edit in the box and change the values by yourself

- I lik… …t. very simple, easy …



Sasha

- for … it a series … ent
- with … esult.
- the … th the environ…
- ric…

Mic… …ative performance

ty…

- tess…
- sou… …ra's textbook
- a …
- Can… …learning

Carl -
inter… …t colors are
-

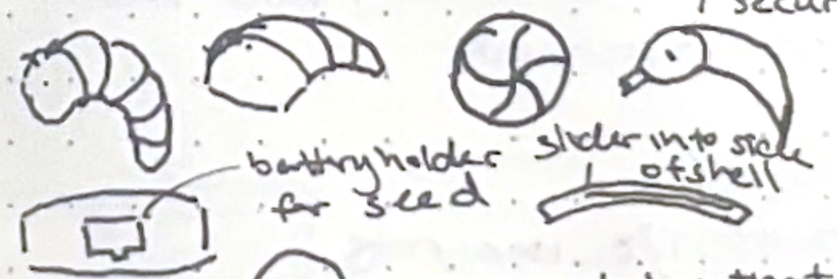
cool

- collects rainwater/water plants

- UI that takes images identify the animal/ instrument & user can play on phone / listen to the animal sounds

Core Crit

- a little hard to use for people
- more type faces
- certain words for different effects (e.g. shatter, blocky)
- drag in your own typeface to use alongside
- testing for GD students
- Kelsey Professor in GD
- type specimen sheet
  - foundaries
- use 'radiation' instead of dsdf
- which ones can you combine (tabs) to ccourage invention of new effects
- user can make own function
- plugin for figma
- font design tool



tomato where there is a book with narrative (or finding Willy) with simple lock system → market it as layers in boxes (outermost to innermost)

AI conversation summarizer → Physical Computation

- in a professional setting, using your phone to record conversation = unprofessionalism
  - survey for professors to confirm
- AI has camera with facial recognition of reading emotions to help tell if someone is actually meaning what they say, or, not (Truth)

→ used essentially in healthcare where therapist need to know if patient is truely feeling

Seat massager inside of Cars

- massage the user inside of the car

Japanese toys sales (mascots) May 23/2024

- adults buying cute toys
- cuteness!
- mascot becoming companions
- consumers of women more than men, but men also take up a significant amount

Appeal
- narrative tied to the mascots
  - people can relate to the mascots
- connect with fans → regular social media post
  - see what workers are doing, and mascot also do
    - = emotional support from toys, not friends etc Japanese don't share
- collaboration with sport/sumo wrestler

Labubu
- surprise character
- customer immerse in world of character
- Intellectual property + sales = 45.5% profit margin

copyright on a 2D image?

seeds

① ledge

ledge

- shell tied that snaps

②

③

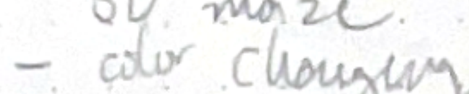
seeds

June 8
7PM night before 7AM

- big ball = lighter material

- manufacturing issue - molds
  - different shapes

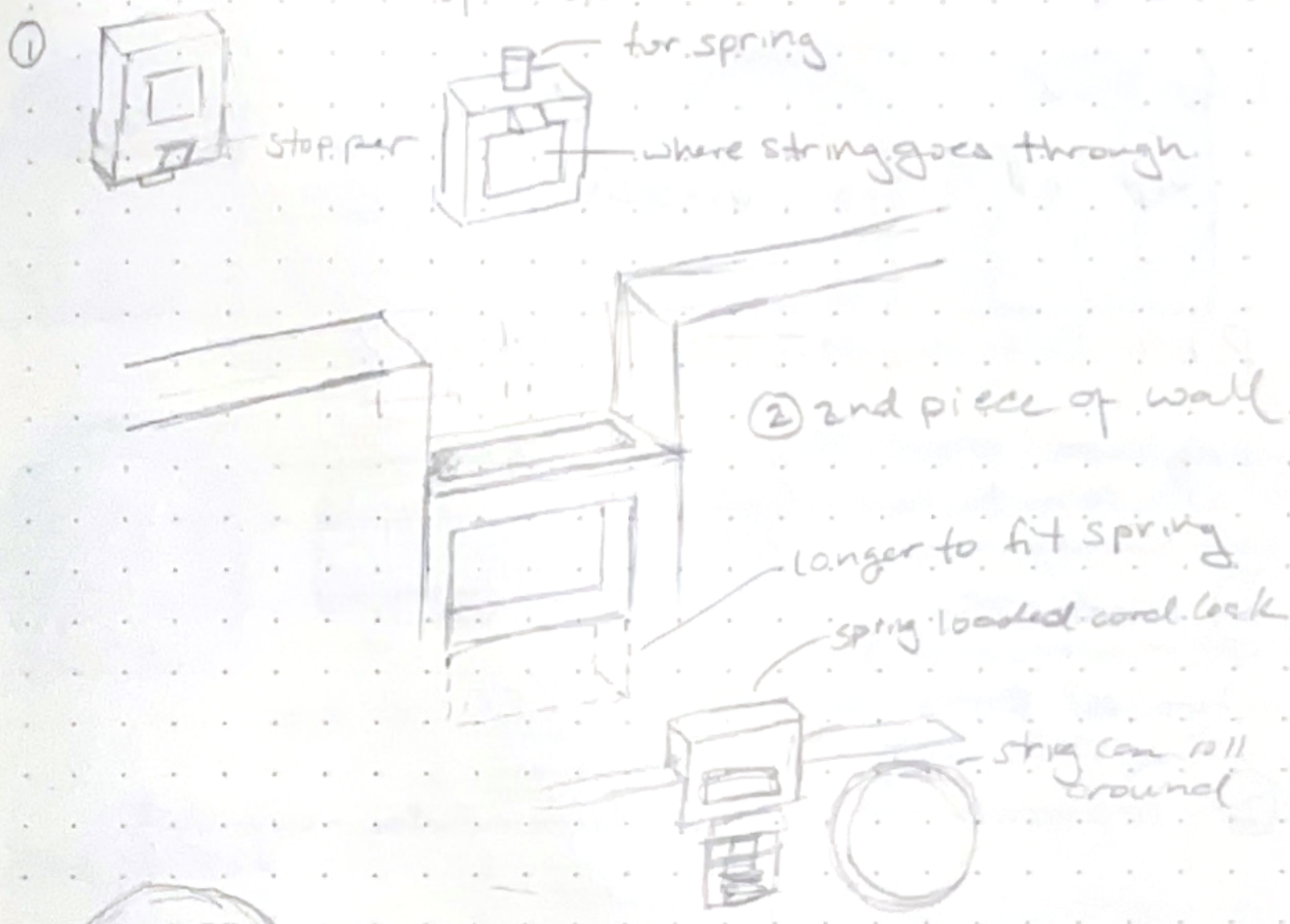
- overlap could provide support
- curved layer

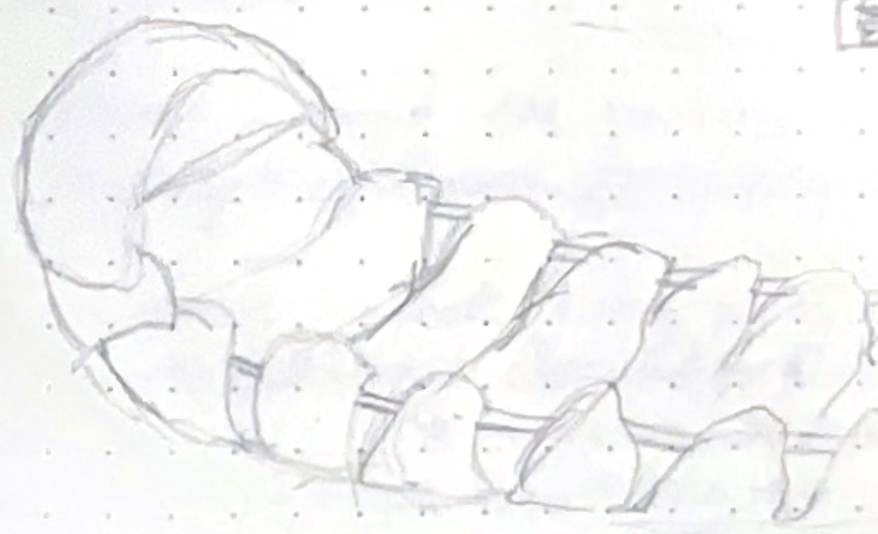


- 3D maze
- color changing

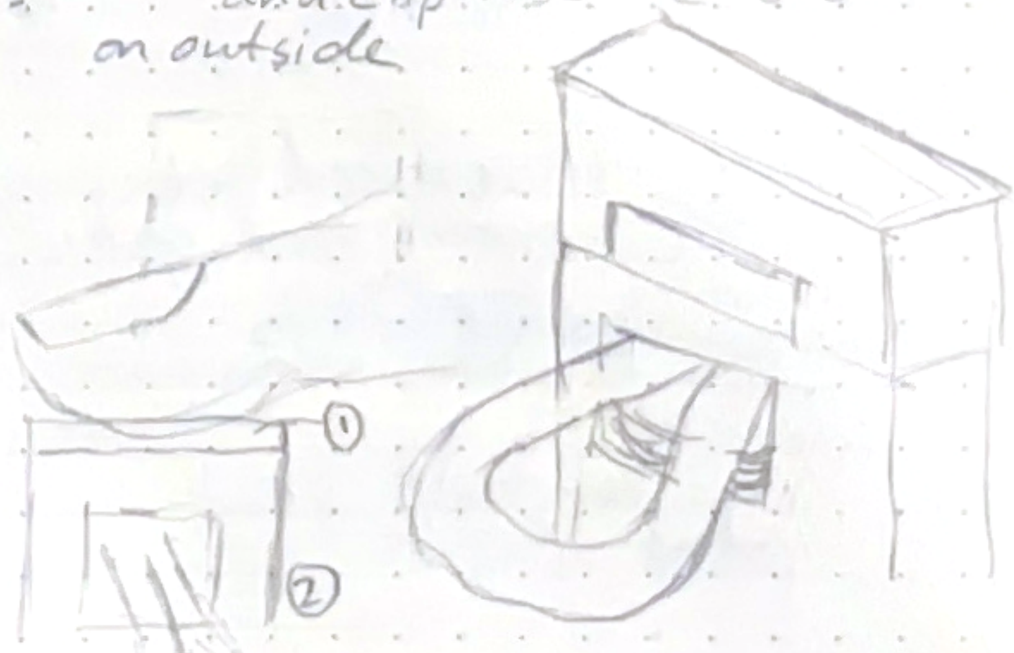
rubber weight material = too heavy
softball < ball > kickball
- semi-transparent, vibrant color change

6/10/2024





indent on outside and clip to secure ball on outside



Brightly

June 10

letter shaker

- seed ball, 10cm, preschool

Robotic Potty toy

- programmed robot that poops (potty tray)
  ↳ nutrition deposit into area & germinate

(I) robot only put out water

- seed germinate = new pattern

(2) pre-programmed image t. certain areas would be watered and germinate

(3) deposit cards onto poppers, bee bot plays certain tune

pattern communicate to kid based on manual

- booklet navel or screen
  ↳ static ↳ animations

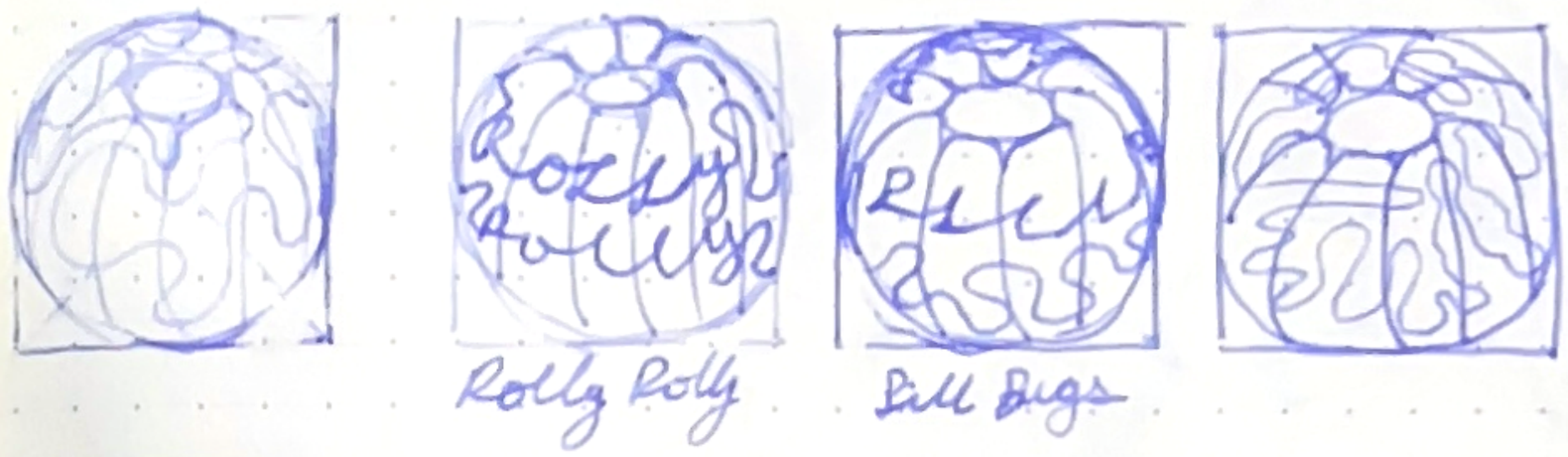
Dorelle

- tile 1 side = seed texture, plant species on other side - memory card game
- research ... ⇒ using light in board games

continuation of attractions

(1) series ↳ plant colony similar to human colony
(2) sunlight shine & shadow on another plant
(3) spatial puzzle plant grow to another

June 11



main issue: ball falling out if it unrolls
① each segment has 1 maze
② a mechanism that prevents the ball from rolling out

June 18/2026

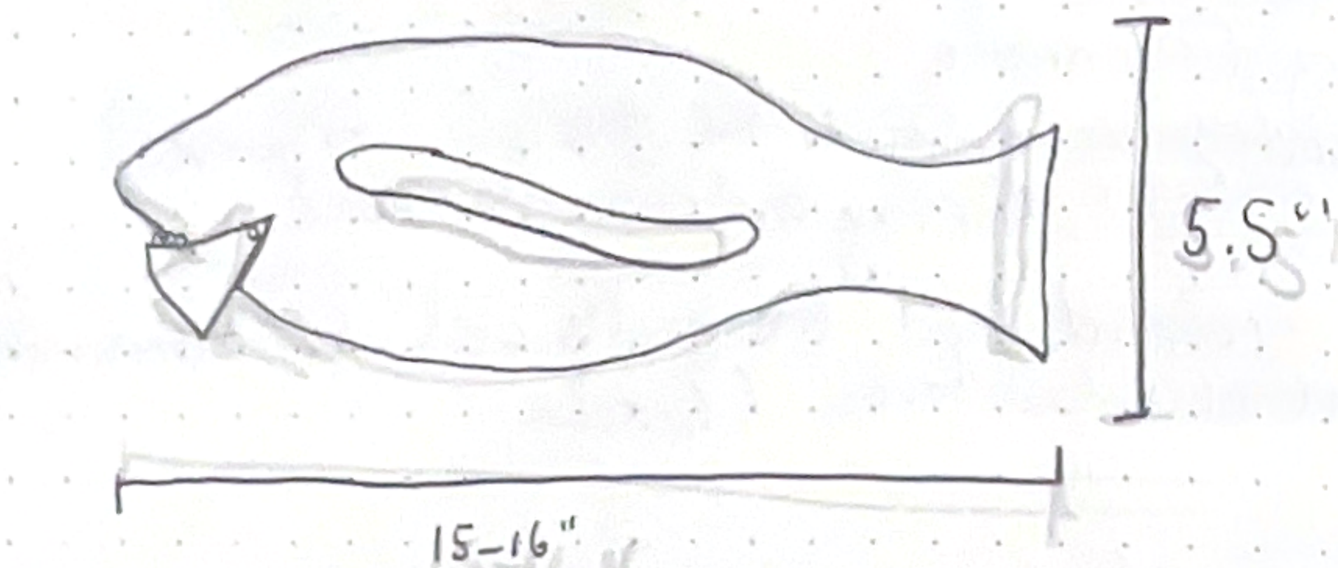
- throw around the ball as bowling (figurines)
  - mystery figurenese + narrative

narrative - what rolly polly eat
- villian characters you can knock down
  ↳ invasive species / predators
- excess rope won't matter or hinge idea

this week
1) narrative
2) figurenes
3) locking mechanism

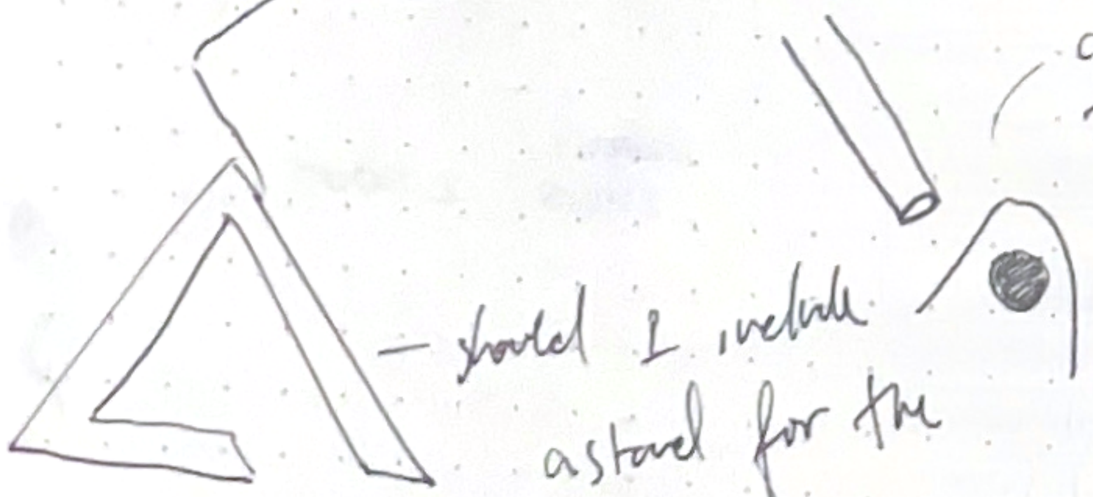
habitat as the hail

↓reveal



- mouths should not be too small to prevent stick being stuck in hole

- the mouth of the fish to small tiny tooth that kids can connect puzzle to missing holes.



- should I include a stand for the fish?

- children like to put things in mouth which can be dangerous and cause a choking hazard

① watering plant app tracker with a soil humidity sensor
- API to weather station
- connects to an app

② humidifier x facial mist / humidifier umbrella

③ Japanese mascot x mental health

④ mental health x fidget toy

⑤ handheld meditation mascot keychain

⑥ keychain soap dispenser → jelly cat

⑦ toilet AI urine monitor

- haptic feedback with vibration/heat
- read biometric (heart beat)

based on heart beat speed set on the app, the vibration controls meditation

press down to PPG heartbeat sensor

big ears indicate listening

- each mascot has a personality to stress, perfectionist etc. things that can be solved with meditation
  ↳ similar to how japanese mascots try to be relatable to its workers audiences (work culture)